

Analysis of DoRA in Multi-Modal Domains

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Background & Motivation

Problem: Full fine-tuning of large pre-trained models (LLMs, ViTs, Wav2Vec) requires updating billions of parameters — prohibitive for most practitioners.

LoRA: Injects trainable low-rank matrices $\Delta W = BA$ into frozen layers. Efficient, but *couples* the magnitude and direction of weight updates.

DoRA Key Insight: Decompose each weight into a learnable **magnitude** scalar m and a **unit-norm direction**, then apply LoRA only to the direction: $W' = \underset{\text{scale}}{m} \cdot \frac{W_0 + BA}{\|W_0 + BA\|_{\text{col}}}$. B initialised to $\mathbf{0}$; m to column norms of W_0 . Overhead vs. LoRA: $+d$ scalars per layer ($< 0.1\%$).

Goals:

- ▶ Reproduce DoRA gains across four modalities
- ▶ Identify when magnitude decoupling matters
- ▶ Compare DoRA vs. LoRA at equal parameter budgets

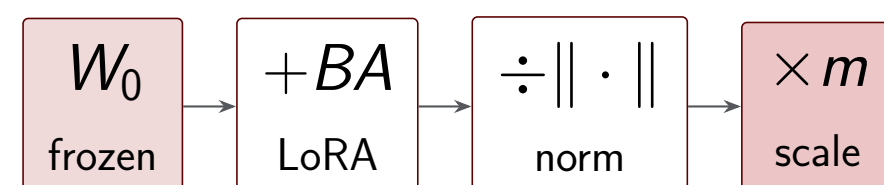
Methodology

Models & Tasks:

- ▶ **NLP:** RoBERTa-base/large on GLUE (SST-2, MRPC, RTE)
- ▶ **Audio:** Wav2Vec2-base on Speech Commands (12 KWS)
- ▶ **Vision:** ViT-B/16 & SigLIP-B/16 on robot grasping
- ▶ **VLA:** Action prediction on Push-T robot task

Shared Config: Rank $r \in \{2, 4, 8, 16, 32\}$; $\alpha = 2r$; target q,k,v,o proj; AdamW + cosine LR; 5 epochs.

DoRA Decomposition Flow:



Implementation: Custom DoRALinear replaces every target nn.Linear in-place; magnitude m , A , B saved separately from frozen base weights.

NLP Results — GLUE Benchmark

MRPC Rank Sweep: (RoBERTa-base; combined F1 + accuracy)

Rank	DoRA (%)	LoRA (%)	Δ
2	87.1	86.2	+0.9
4	86.6	87.7	−1.1
8	87.8	85.9	+1.9
16	90.3	87.3	+3.0
32	89.8	90.1	−0.3

DoRA at $r = 16$ (90.3%) beats LoRA's best ($r = 32$: 90.1%) at **half the parameters**.

Cross-Task at $r = 8$: (RoBERTa-base; best checkpoint)

Task	DoRA	LoRA	Δ
SST-2 (Acc)	93.1%	93.3%	−0.2
MRPC (F1)	90.7%	90.1%	+0.6
RTE (Acc)	71.1%	70.8%	+0.3

DoRA gains on **low-data tasks**: +0.3pp RTE, +0.6pp MRPC F1. On large-data SST-2, LoRA converges equally well and magnitude decoupling provides no additional benefit.

SST-2 "...crowd-pleasing → **Positive**
as a documentary
can get."

MRPC S1: "Corixa rose → **Not Equiv.**
54c to \$7.74." S2:
"...rose 54c, or 8%,
to \$7.74."

Real Validation Samples:

RTE P: "Leaders signed → **Not Entailed**
Egypt–Israel treaty
in 1979." H:
"Treaty signed in
1979."

Audio Results — Speech Commands

Wav2Vec2-base (67M params) fine-tuned on Google Speech Commands v0.02 — 84.8k utterances, **12 keyword classes**.

Method	Val Acc	Test Acc	Trainable	Time
DoRA ($r = 8$)	98.6%	89.7%	826.6k	152.8 min
LoRA ($r = 8$)	98.5%	89.0%	789.8k	190.8 min

+0.7% test accuracy, 25% less training time. Faster directional gradient convergence on the audio encoder.

Vision & Robot Results

Cornell Grasp: Success rate ($\text{IoU} \geq 0.25$ & $|\Delta\theta| \leq 30^\circ$):

Backbone	DoRA	LoRA	Full FT
ViT-B/16	7.9%	6.2%	0.6%
SigLIP-B/16	19.8%	20.3%	15.8%

SigLIP ($3\times$ over ViT) dominates; DoRA $\approx$ LoRA. Full FT collapses on 885 images.

Qualitative: SigLIP + DoRA: Green = predicted Red = ground truth



Push-T VLA Task: DoRA and LoRA applied to a 126-layer VLA for robotic push-block manipulation. Action MSE after 3 epochs: DoRA 63.6 → 27.9; **LoRA 62.1 → 24.8** (LoRA outperforms — too few epochs for magnitude decoupling to take effect).



DoRA adapter driving Push-T rollouts

Conclusions & References

- ▶ **Low-data NLP:** +0.3 pp RTE, +0.6 pp MRPC F1 (both $r = 8$); DoRA at $r = 16$ matches LoRA at $r = 32$ with half the parameters
- ▶ **Audio:** +0.7% test accuracy, 25% faster training on Wav2Vec2 + Speech Commands
- ▶ **Vision:** SigLIP benefits more than ViT; gains are modality- and architecture-dependent
- ▶ Gains **diminish on large models** where LoRA saturates
- ▶ DoRA is a **drop-in LoRA replacement** across all four modalities tested

[1] Liu et al. (2024). DoRA: Weight-Decomposed Low-Rank Adaptation. *ICML 2024*.

[2] Hu et al. (2022). LoRA. *ICLR 2022*.

[3] Wang et al. (2018). GLUE Benchmark. *EMNLP*.

[4] Warden (2018). Speech Commands. *arXiv*.